Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative disturbance (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (102 sites, 362 site-years) · deadtrees v4.1

ΔRMSE = RMSE(without D) − RMSE(with D)

	XGBoost (Optuna) · 12m	XGBoost (Optuna) · 24m	LightGBM (Optuna) · 12m	LightGBM (Optuna) · 24m
GPP (g C m⁻² season⁻¹)	59% improved	63% improved	60% improved	66% improved
NEP (g C m⁻² season⁻¹)	59% improved	64% improved	60% improved	61% improved
ET (mm season⁻¹)	38% improved	45% improved	52% improved	47% improved
WUE (g C mm⁻¹)	55% improved	66% improved	60% improved	61% improved

Relative disturbance (%)
site maximum, 500 m
breaks 12.0 | 22.8

60
40
20
0

Sites (sorted by ΔRMSE within each panel)